

Castalia

A four-channel electrochemical sensing SoC with a five-hart RISC-V back end

Maxx Seminario

University of Nebraska–Lincoln

September 20, 2026

Why ovarian: late, lethal, and unscreened

Castalia

Maxx Seminario

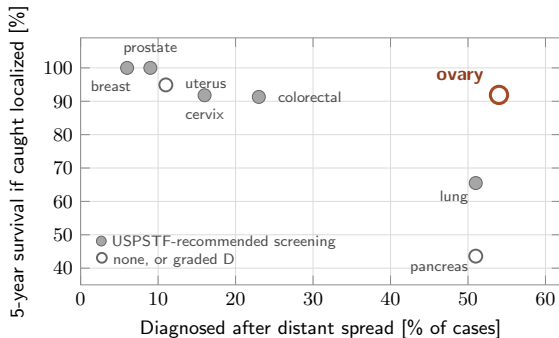
Target application

The chip

Analog front end

From currents to a score

References



Site	Distant at dx	Deaths/100 cases	Screening (USPSTF)
Ovary	54 %	59	D
Pancreas	51 %	78	D
Lung	51 %	54	B, LDCT
Colorectal	23 %	35	A
Cervix	16 %	31	A
Uterus	11 %	21	none
Prostate	9 %	11	C, PSA
Breast	6 %	13	B

[1] SEER, 2026 estimates; [2, 3] USPSTF

Most lethal gynecologic cancer [1]: 92 % survivable if caught localized, yet 54 % present after spread, and screening is graded D [2]. UKCTOCS moved stage, not mortality [4].

One serum aliquot, four sites, one score

Castalia

Maxx Seminario

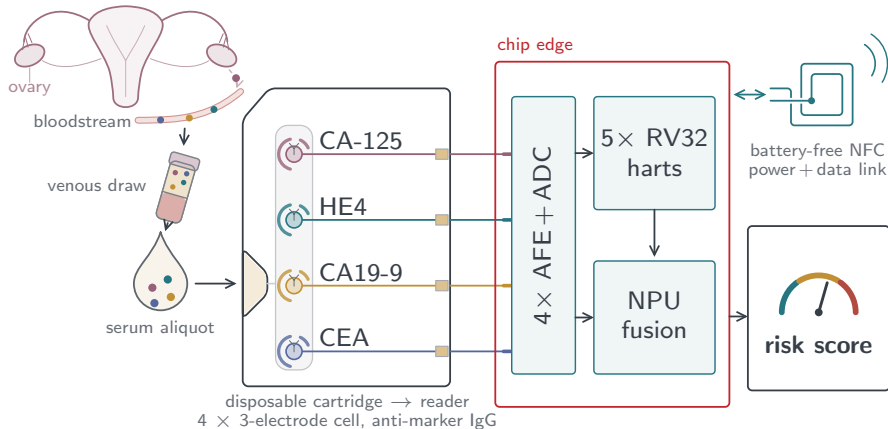
Target application

The chip

Analogue front end

From currents to a score

References



Symptomatic-patient triage on a disposable four-site cartridge; four markers read at the same instant, fused on chip.

No single ovarian marker is a usable test [5]

Castalia

Maxx Seminario

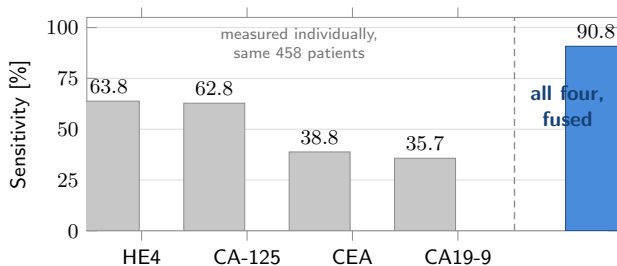
Target
application

The chip

Analog front end

From currents to
a score

References



Panel	Sensitivity
Best single (HE4)	63.8 %
Four fused	90.8 %
ROMA, FDA-cleared [6]	2 markers
This chip	4 channels

Clinical data (458 patients), not simulation. The decision is a learned combination, so the unit of integration is four concurrent channels plus the processor fusing them.

Four markers resolved in one differential-pulse sweep

Castalia

Maxx Seminario

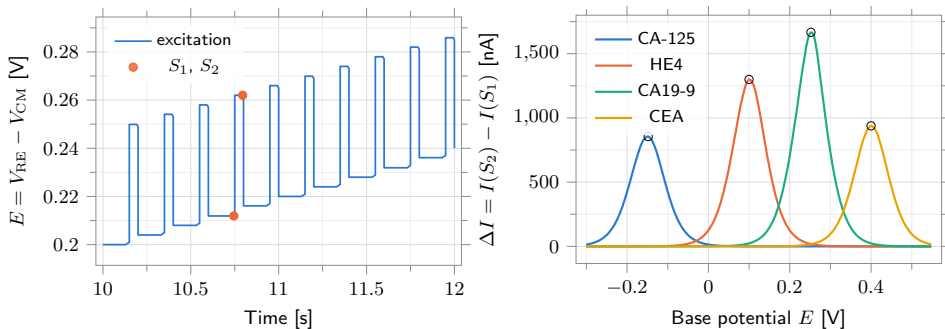
Target
application

The chip

Analog front end

From currents to
a score

References



■ schematic sim, pre-layout

Peak position identifies the reporter, peak height carries concentration. Apexes within 2 mV and 1 % of the assay targets; blank below 1 pA.

Every clinical cut-off lands inside the channel window

Castalia

Maxx Seminario

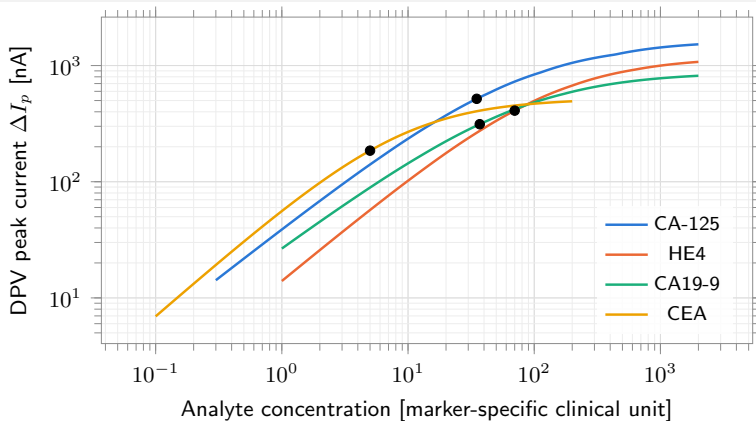
Target application

The chip

Analog front end

From currents to a slope

References



■ schematic sim,

pre-layout

CA-125 35 U/mL, HE4 70 pmol/L, CA19-9 37 U/mL, CEA 5 ng/mL: all four on the steep part of the curve at one R_f setting.

Castalia at a glance

Castalia

Maxx Seminario

Target
application

The chip

Analog front end

Front currents to
a scan

References

Digital

Harts	5 (1 orchestrator, 4 channel tiles)
ISA	rv32imac_zba_zbb_zbs_zbc
Clock	24 MHz rated, closed at 25 MHz
Measured CPI	1.260 aggregate, nine benchmarks
Private TCM	8 KiB per hart, 1-cycle
Shared RAM	80 KiB behind a round-robin arbiter
Synchronisation	16 hardware mutexes, LR/SC, AMOs
Interrupts	claim/complete, any vector to any hart
Accelerator	NPU: MLP, conv-1D, XNOR, GEMM
Power domains	4 MTCMOS, one per channel tile
Debug	JTAG, RISC-V Debug Module
Field boot	NFC ISO 14443A harvested boot

Analog and physical

Measurement sites	4 identical, simultaneous
Topology	bipolar grounded-WE potentiostat
Electrode pads	12 (CE/RE/WE $\times$ 4)
Converter	10-bit SAR per channel, extracted
Reference DAC	12-bit R-2R per channel
Feedback resistor	$18\text{ k}\Omega + 6\text{ k}\Omega \cdot N$, 320 codes
Full scale	$\pm 40.7\text{ }\mu\text{A}$ to $\pm 0.41\text{ }\mu\text{A}$
EIS	firmware mode of the same channels
Analog supply	2.5 V , $v_{\text{cm}} = 1.25\text{ V}$
Process	65 nm CMOS, 1P8M, HVT cells
Die / package	$3.0 \times 3.0\text{ mm}$ / LQFP-100
Digital power	10.72 mW post-route, SS, 25 MHz

Digital numbers from the hardened flow; analog numbers schematic-level, except the SAR, which is extracted.

Block diagram

Castalia

Maxx Seminario

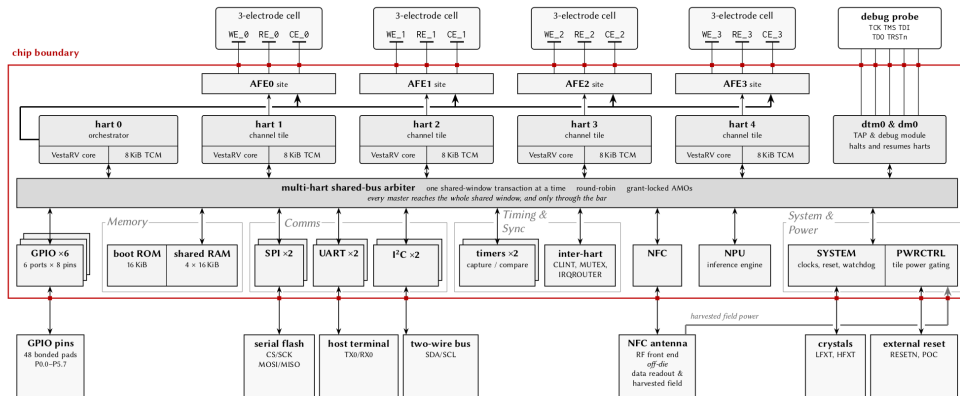
Target
application

The chip

Analogue front end

From currents to
a score

References



Four channel tiles own four AFE sites; every master reaches ROM, RAM, NPU and peripherals only through the arbiter.

Physical implementation

Castalia

Maxx Seminario

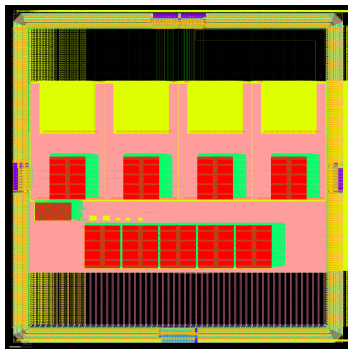
Target
application

The chip

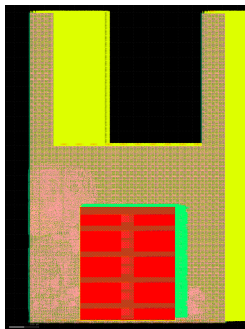
Analog front end

From currents to
a score

References



chip top, 3.0 × 3.0 mm



one hardened channel tile

Process	65 nm, 1P8M, HVT
Core box	2690 × 2690 μm
Tile	660 × 880 μm
Tile logic	≈98 k NAND2-eq.
Tile SRAM	8 KiB
Package	LQFP-100, 14 × 14 mm

■ hardened flow, post-route

One tile hardened once, instantiated four times; the notch in each tile is the analog channel's keep-out.

Digital, as closed

Castalia

Maxx Seminario

Target
application

The chip

Analog front end

From currents to
a score

References

Whole chip, post-route

Internal	3.78 mW
Switching	2.33 mW
Leakage	4.61 mW
Total	10.72 mW

One channel tile

Total	1.761 mW
Gated	0

Measured CPI, timed kernels

Benchmark	Instr.	CPI
spmv	802 447	1.143
multiply	20 892	1.163
vvadd	2 408	1.374
median	4 015	1.398
qsort	123 205	1.398
rsort	146 548	1.412
dhrystone	200 523	1.504
towers	3 747	1.682
memcpy	11 017	1.726
Aggregate	1 314 802	1.260

Timing at 25 MHz

reg2reg WNS	+99 ps
Leakage share	43 %

■ hardened flow, post-route

SS 0.9 V 125 °C, statistical activity 0.2. Gating one tile removes 1.76 mW; the always-on band pays nothing.

Grounded-WE two-amplifier potentiostat

Castalia

Maxx Seminario

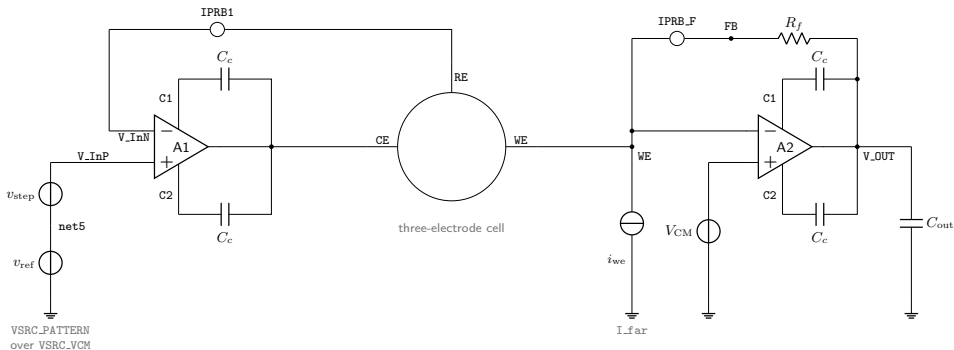
Target
application

The chip

Analog front end

From currents to
a scope

References



A1 drives CE with RE in feedback; A2 holds WE at v_{cm} through R_f . $v_{out} = v_{cm} - I_{we}R_f$, both polarities.

Bipolar accuracy through zero

Castalia

Maxx Seminario

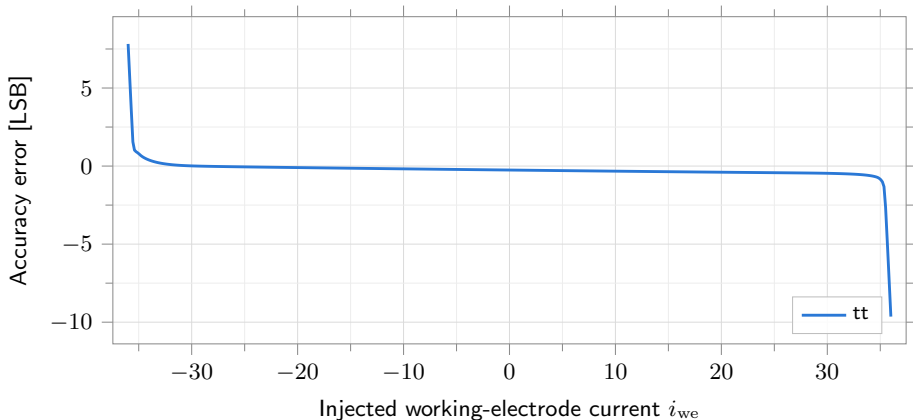
Target
application

The chip

Analog front end

From currents to
a score

References



■ schematic sim, pre-layout

No kink at the crossing: class-AB output, no dead zone. This is the measurement rev-1's PMOS mirror could not make.

Cyclic voltammetry through the assembled channel

Castalia

Maxx Seminario

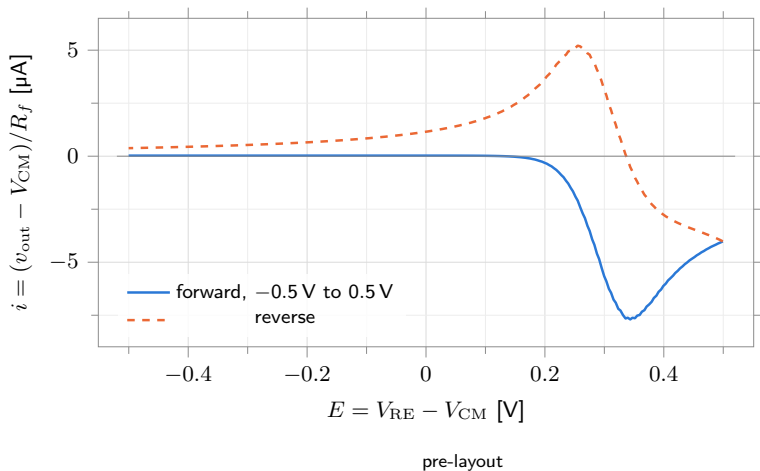
Target
application

The chip

Analog front end

From currents to
a scope

References



Staircase potential from the on-chip DAC, current digitised by the on-chip SAR, unclipped over the sweep.

Programmable feedback resistor: one string, 320 codes

Castalia

Maxx Seminario

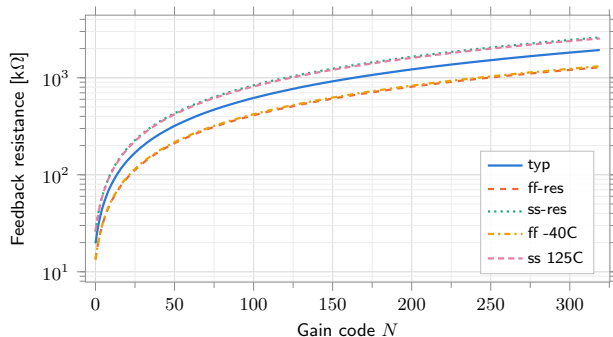
Target application

The chip

Analog front end

From currents to a scale

References



$$R_f = 18 \text{ k}\Omega + 6 \text{ k}\Omega \cdot N$$

N	R_f	Full scale
0	19.6 kΩ	±40.7 μA
63	397 kΩ	±2.02 μA
319	1.93 MΩ	±0.41 μA
LSB at 319		0.824 nA

Strictly monotonic at every corner, including the four thermometer seams. Maximum gain is noise-limited, not quantisation-limited.

10-bit SAR ADC, extracted netlist

Castalia

Maxx Seminario

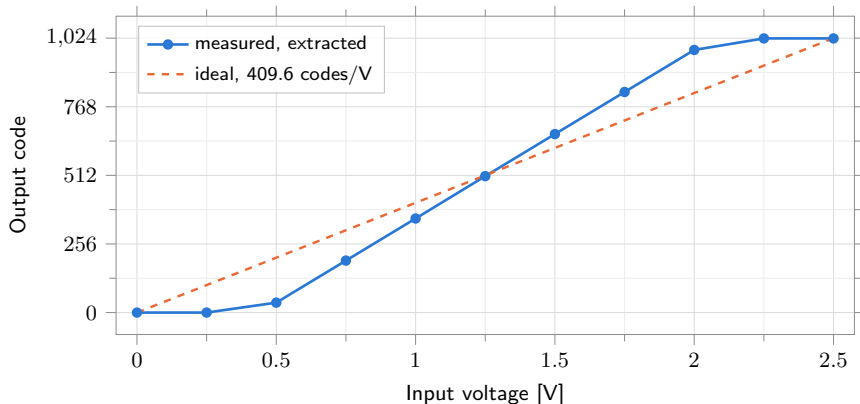
Target
application

The chip

Analog front end

From currents to
a scope

References



■ extracted

netlist

23 clocks per conversion, 870 kSa/s, 50 pJ. The capacitor array exists only in layout, so this is the one block characterised extracted.

Impedance spectroscopy on the same channel

Castalia

Maxx Seminario

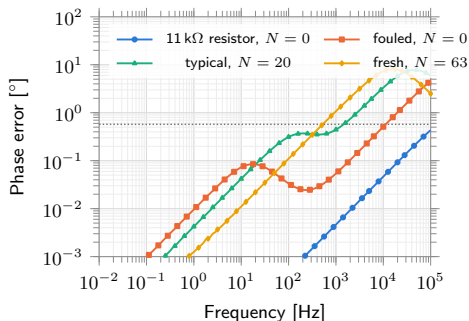
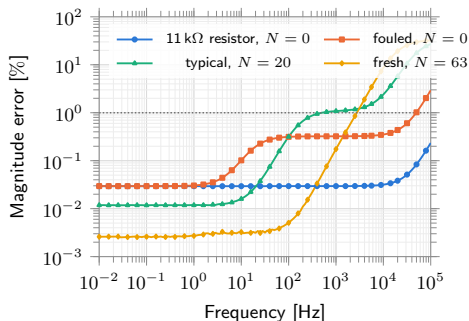
Target application

The chip

Analog front end

From currents to a scale

References



Excitation	Demodulation	Accuracy	$ Z $ range
2-code DAC square	firmware, channel hart	$\pm 1\%$, $\pm 0.6^\circ$ to 5.05 kHz; $\pm 5\%$ to 26.5 kHz	246 Ω – 12.1 M Ω

■ schematic sim, pre-layout

No EIS engine. Impedance is a ratio of two readings through one converter; the DAC drops out of the accuracy budget.

One AFE channel, as simulated

Castalia

Maxx Seminario

Target application

The chip

Analog front end

From currents to a scalar

References

Parameter	Value	Condition
CE compliance	13.2 mV – 2.480 V	2.5 V rail, corners
Cell current delivered	$\pm 52.5 \mu\text{A}$	12 k Ω cell
TIA compliance	$\pm 0.80 \text{ V}$ about v_{cm}	
Feedback resistor	19.6 k Ω – 1.93 M Ω , 320 codes	monotonic at every corner
Full-scale current	$\pm 40.7 \mu\text{A}$ ($N=0$) to $\pm 0.41 \mu\text{A}$ ($N=319$)	SAR window $\cap$ TIA compliance
Current quantum	0.824 nA at maximum gain	below the 2 nA noise budget
Bipolar accuracy	0.25–0.38 LSB worst	560 k Ω , 1 M Ω electrode, corners
Control-loop PM	79–80°	five corners, into a cell
TIA-loop PM	89–93°	12-cell electrode grid
Input current noise	341 pA standalone, 3.6–21 nA into an electrode	0.1–100 Hz, 560 k Ω
Zero-current offset	$\sigma(v_{\text{re}}) = 5.72 \text{ mV}$, gain-independent	200 MC samples, calibratable
Quiescent current	197–250 μA per channel	amplifiers + local bias

■ schematic sim, pre-layout ■ Monte Carlo, 200 samples

Pre-layout, electrode held nominal: no row includes electrode-to-electrode spread.

The fusion runs on the NPU

Castalia

Maxx Seminario

Target
application

The chip

Analog front end

From currents to
a score

References

$$s = \sigma\left(\sum_{k=1}^4 w_k \hat{c}_k + b\right)$$

Input	peak current, peak potential, baseline $\times 4$
Mode	0, dense layer, one sigmoid
Output	one risk score
Weights	trained off-chip, staged by firmware

Per site

Hart	1, ownership enforced in hardware
Potential	own 12-bit DAC
Converter	own 10-bit SAR
Power	own MTCMOS domain

The logistic form of the cleared two-marker algorithm, extended to four inputs and moved from the central laboratory onto the electrode.

References

Castalia

Maxx Seminario

Target
application

The drip

Analgesic front end

From currents to
a score

References

- [1] National Cancer Institute, *SEER Cancer Stat Facts*, per-site pages, <https://seer.cancer.gov/statfacts/>, accessed 20 Sep. 2026. Stage and 5-year relative survival: SEER 21, 2016–2022; cases and deaths: 2026 estimates.
- [2] D. C. Grossman *et al.*, “Screening for ovarian cancer: US Preventive Services Task Force recommendation statement,” *JAMA*, vol. 319, no. 6, pp. 588–594, 2018.
- [3] US Preventive Services Task Force, *A and B Recommendations; Prostate Cancer: Screening*, 2018 (C); *Pancreatic Cancer: Screening*, 2019 (D).
<https://www.uspreventiveservicestaskforce.org/>.
- [4] U. Menon *et al.*, “Ovarian cancer population screening and mortality after long-term follow-up in the UK Collaborative Trial of Ovarian Cancer Screening (UKCTOCS),” *The Lancet*, vol. 397, pp. 2182–2193, 2021.
- [5] F. Chen *et al.*, “Clinical analysis of four serum tumor markers in 458 patients with ovarian tumors: diagnostic value of the combined use of HE4, CA125, CA19-9, and CEA,” *Cancer Manag. Res.*, vol. 10, pp. 1313–1318, 2018.
- [6] R. G. Moore *et al.*, “A novel multiple marker bioassay utilizing HE4 and CA125 for the prediction of ovarian cancer in patients with a pelvic mass,” *Gynecol. Oncol.*, vol. 112, no. 1, pp. 40–46, 2009.